20/7/25

HW5 Assignment

1) Given,

Multivariate random normal vector. $X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix}$

Covariance matrix Σ & Precision Matrix $Q = \Sigma^{-1}$

a) Are X_3 & X_4 correlated ~~correlated~~?

$\Sigma_{3,4} = 0$ & $\Sigma_{4,3} = 0$

So, the covariance is 0.

$\Rightarrow X_3$ & X_4 are uncorrelated //

b) Are X_3 & X_4 conditionally correlated given others.

Conditional correlation is determined by precision matrix Q.

$X_i \perp X_j \mid \text{others} \Leftrightarrow Q_{ij} = 0$

Check $Q_{3,4} \Rightarrow 0$.

$\Rightarrow X_3$ & X_4 are conditionally independent //

c) Markov blanket of X_2?

Markov blanket of X_i is the set of variables with nonzero entries in the i^{th} row/column of precision matrix Q (off diagonal)

Row 2 of Q:

$Q_{2,\cdot} = [3, 5, 0, 0]$

Non-zero off diagonal entries are with Q_{21} only

~~Therefore Markov blanket of X_2 is $\{X_1\}$~~ //

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d) Covariance matrix of

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} X_1 + X_4 \\ X_2 - X_4 \end{bmatrix}$$

To find $\Sigma_Y = \begin{bmatrix} Var(Y_1) & Cov(Y_1, Y_2) \\ Cov(Y_2, Y_1) & Var(Y_2) \end{bmatrix}$

$Var(Y_1) = \Sigma_{11} + \Sigma_{44} + 2\Sigma_{14} = 0.71 + 0.2 + 0$
$= 0.91$

$Var(Y_2) = \Sigma_{22} + \Sigma_{44} - 2\Sigma_{24} = 0.46 + 0.2 + 0$
$= 0.66$

$Cov(Y_1, Y_2) = Cov(X_1, X_2) - Cov(X_1, X_4) + Cov(X_4, X_2) - Cov(X_4, X_4)$
$= 0.43 - 0 + 0 - 0.2$
$= -0.63$

$$\Rightarrow \Sigma_Y = \begin{bmatrix} 0.91 & -0.63 \\ -0.63 & 0.66 \end{bmatrix}$$

——— X ———

2) Given
Mixture of 2 Gaussians, each with unit variance,

$$f(x) = \frac{1}{2}\phi(x|\mu_1, 1) + \frac{1}{2}\phi(x|\mu_2, 1)$$

$x^{(1)} = -1, \; x^{(2)} = 1$

a) One EM update starting from $\mu_1 = -2$, $\mu_2 = 2$

We know,

$$\gamma_i = \frac{\phi(x^{(i)}|\mu_1, 1)}{\phi(x^{(i)}|\mu_1, 1) + \phi(x^{(i)}|\mu_2, 1)}$$

$x^{(1)} = -1$; closer to $-2 \Rightarrow$ high γ for μ_1
$x^{(2)} = 1$; closer to $2 \Rightarrow$ high γ for μ_2

$$\gamma_1 = \frac{1}{1+e^{-4}} \approx 0.98, \Rightarrow \gamma_2 = \frac{1}{1+e^{4}} \approx 0.02$$

$$\Rightarrow \mu_1^{new} = \frac{\gamma_1 x^{(1)} + \gamma_2 x^{(2)}}{\gamma_1 + \gamma_2} = \frac{(-1)(0.98) + (1)(0.02)}{0.98 + 0.02}$$

$$\mu_1^{new} \approx -0.96$$

$$\mu_2^{new} = \frac{(1-\gamma_1)x^{(1)} + (1-\gamma_2)x^{(2)}}{(1-\gamma_1)+(1-\gamma_2)}$$

$$= \frac{(-1)(0.02) + (1)(0.98)}{0.02 + 0.98} = 0.96$$

$$\Rightarrow \mu_2^{new} \approx 0.96$$

b) Will EM converge to $\mu_1 = -1$ & $\mu_2 = 1$?

The data points -1 & $+1$ are symmetric. But EM updates the means to be closer to clusters of weighted points — not necessarily the exact points.

So, they will ~~convergenge~~ converge closer to the data but not exactly equal to them unless initialized that way

c) Fixed point if $\mu_1 = \mu_2 = 2$ for EM.

If $\mu_1 = \mu_2 = 2$, then both components are identical Gaussians.

So, the E-step gives,

$\gamma = 0.5$ for both points & both components

Then M-step updates both means as,

$$\mu_1^{new} = \mu_2^{new} = \frac{1}{2}\left(x^{(1)} + x^{(2)}\right) = \frac{-1+1}{2} = 0.$$

For next iteration, still identical $\Rightarrow$ update stays at 0.

So, Fixed point is,

$$\mu_1 = \mu_2 = 0$$

d) Fixed ~~for~~ point from, $\mu_1 = -2$, $\mu_2 = 2$ for K-means.

K-means assigns each point to closest center,

$x^{(1)} = -1 \rightarrow$ closer to $\mu_1 = -2$
$x^{(2)} = 1 \rightarrow$ closer to $\mu_2 = 2$

$\Rightarrow$ Each cluster has 1 point, hence,

$\mu_1^{new} = -1$, $\mu_2^{new} = 1$

This is stable for next assignments too, it wont change,

So, Fixed points for K-means are,

$\mu_1 = -1$, $\mu_2 = 1$